

CHE 253M – Measurement, Control and Data Analysis Laboratory Spring 2026

Description: This laboratory course is designed to provide hands-on experience with fundamental chemical engineering measurements, the demonstration of chemical engineering principles, and the experimental analysis and solution of chemical engineering problems. Specifically, the experiments cover temperature and pressure measurements, gas and liquid flow measurements, heat transfer, laminar and turbulent pipe flow, frictional pressure losses, qualitative feedback process control, and statistical data analysis. Your team will conduct four experiments over the course of the semester. In these experiments, you will be required to propose solutions to scale-up and design problems based on your experimental results.

Learning Objectives: By the completion of this course, students will be able to:

- Design experimental plans and perform laboratory experiments logically.
- Apply critical thinking, theoretical models, and chemical engineering principles to analyze experimental data.
- Verify experimental deviations from theoretical models utilizing chemical engineering principles.
- Evaluate the direct application of engineering concepts in the solution of real-world problems.
- Summarize experimental findings, create compelling discussions, and propose practical engineering solutions.

By the practice of these objectives, students will be able to:

- Develop written and oral technical communication skills
- Demonstrate and practice laboratory safety
- Work effectively in teams while enhancing individual teaming skills

Materials: Canvas will be your main source of materials for the course. Laboratory manuals and other information will be posted, as well as links to other sources of information.

Class: M, T, W or Th; 2:00 – 6:00 pm. Laboratory located in CPE 1.420

Instructors: Professor Carlos Landaverde Alvarado

Professor Joan F. Brennecke

Office Hours:

Professor Landaverde: Questions? Please stop by CPE 1.480 (Senior lab) on M, T, W from 4:00 pm to 5:00 pm, or you can email me for an appointment if those times don't work for you.

Professor Brennecke: By appointment; please email.

TAs: Available for office hours during times listed in the laboratory schedule. Please note office hours are mandatory on the week following each laboratory experiment.

Grades:

Your grade in the course will be based on four components:

- 4% Completion of safety training. This involves three UT training courses (OH101, OH102 and OH202, or OH238) and two SChE courses (ELA954 and ELA950). These must be completed to be allowed to work in the lab. The deadline for completion to receive credit is 2:00 pm on January 20 (Tuesday Section), January 21 (Wednesday Section), January 22 (Thursday Section), or January 26 (Monday Section). Further information about this requirement is detailed in section 3 below.
- 90% There are four laboratory experiments, which will be given equal weight. Components for each experiment include a pre-lab presentation and a pre-lab report (containing objectives, methods, a safety evaluation and sample calculations), both of which are completed prior to conducting the experiment, and the final report. The final report will be a full laboratory report for the Mixing experiment and a technical memo for the Liquid Flow, Gas Flow, and Heat Transfer experiments.
- 5% Participation and Laboratory Safety. Active engagement in planning and conducting the laboratory experiments and analyzing the results is a critical part of this course. Laboratory safety is a very important part of your professional development as a chemical engineer. All group members are expected to fully participate and to follow the safety rules of the laboratory. Participation and safety will be judged primarily by the instructors and TAs. However, you will be required to fill out a team peer evaluation twice during the course of the semester.
- 1% Attending lecture and office hours. There will be 3 required lectures and 4 required office hours. Required office hour attendance for each team is listed in the laboratory schedule.

The full range of available grades (including + and -) will be utilized. Grades may be curved as necessary.

A	A-	B+	B	B-	C+	C	C-	D	F
93.00-100%	90.00-92.99%	87.00-89.99%	83.00-86.99%	80.00-82.99%	77.00-79.99%	73.00-76.99%	70.00-72.99%	60.00-69.99%	<60.00%

Participation and Laboratory Safety: “Hands on experience” is a critical component of this course. It is for this reason that a portion of the entire grade is based upon how effectively you participate in the laboratory experience. In general, it is expected that you will be fully engaged in the planning and running of the experiment. This also includes following the appropriate safety procedures and rules while completing your laboratory experiments.

- This means that you must attend your group’s four laboratory sessions. University-recognized excuses are required if you are unable to do so. University-recognized excuses include medical issues requiring a doctor visit and/or hospitalization, religious holy days, military service, and qualifying medical or military family leave. Please request a class absence notification from the Office of the Dean of Students in these cases. Requests can be submitted [here](#). Moreover, you need to contact the instructor (email) as soon as the University-recognized excuse is known and, obviously, BEFORE the laboratory period. Job interviews, extracurricular activities, etc. are not approved excuses.
- The best way to ensure active participation is through preparation. You will need to study the laboratory manual and have done sample calculations and the safety section before the

lab period so that you know exactly what you are going to do in the lab. You need to have planned out what needs to be done in the lab before the lab period.

- You cannot let one or more group members dominate the group. Everyone needs to speak up about what needs to be done, what is the next step, etc. The instructors and TA may ask questions to try to encourage participation by all group members. However, it is each of your responsibility to make sure that you actively contribute.
- Peer evaluations will also represent a percentage of your final participation grade and will play a role in your final course grade. Students with consistently low participation grades will be contacted by the instructors throughout the semester to pursue strategies to improve their participation and engagement in the course.

Course Organization:

- Experiments will be performed in teams (designated as the RED, ORANGE, YELLOW, GREEN, BLUE, and PURPLE groups). Most teams will be 4 students. Teams will be assigned by the instructors. Given specific enrollment numbers for each day, some groups may be 3 students.
- All work, besides performance of the actual experiment, preparing the pre-lab report (containing objectives, methods, a safety section, and sample calculations), and preparing/presenting the pre-lab presentations, will be done individually. **Laboratory reports are individual.**
- Over the course of the semester, each group will do each of the 4 experiments.
- In-lab experiments will begin on Tuesday, January 20 and conclude on Monday, April 13.
- The required introductory lectures will be on the following dates at 2:00 pm in CPE 1.418:
 1. Monday Section: Monday, January 12.
 2. Tuesday Section: Tuesday, January 13.
 3. Wednesday Section: Wednesday, January 14.
 4. Thursday Section: Thursday, January 15.
- There will be two additional required class lectures that will be held at 2:00 pm on the following dates in CPE 1.418:

Lecture 2:

1. Tuesday Section: Tuesday, February 24.
2. Wednesday Section: Wednesday, February 25.
3. Thursday Section: Thursday, February 26.
4. Monday Section: Monday, March 2.

Lecture 3:

1. Tuesday Section: Tuesday, April 21.
 2. Wednesday Section: Wednesday, April 22.
 3. Thursday Section: Thursday, April 23.
 4. Monday Section: Monday, April 27.
- Experiments #1 and #2 will be: Mixing/Statistical Process Control and Liquid Flow.
 - Experiments #3 and #4 will be: Heat Transfer and Gas Flow.

Laboratory Safety:

- You must complete the five online safety training modules (three UT and two SACHÉ) before 2:00 pm on January 20 (Tuesday Section), January 21 (Wednesday Section),

January 22 (Thursday Section), or January 26 (Monday Section) to **receive credit** for having completed the modules.

- Please note that if you do not complete the three UT safety courses before **2 pm** on the first day your group is scheduled to do an experiment, you will not be allowed to run the experiment due to university Environmental Health & Safety (EH&S) policies.
- Long pants and closed-toe shoes are required. **Students who fail to wear long pants will receive a 0/10 grade in safety, students wearing open-toe shoes will not be allowed in the lab. You must also wear a lab coat, which will be supplied by the department.**
- You must wear disposable gloves, supplied by the department, at all times in lab.
- Your final laboratory grade also depends on following the in-lab safety rules and policies.

Laboratory Deadlines:

Deadline	Individual/Group	Task
January 20 (Tuesday Section), January 21 (Wednesday Section), January 22 (Thursday Section), or January 26 (Monday Section)	Individual	Complete and upload certificate of completion of the five online safety training modules (three UT and two SACHe)
2 pm the day your group is to run an experiment	Group	Complete the pre-lab report presentation
2 pm the day your group is to run an experiment	Group	Submit pre-lab report
One week after your group runs an experiment	Individual	Attend required office hours
One week after your group runs an experiment	Individual	Submit REPORT APPENDIX by 11:59 pm
Two weeks after your group runs an experiment	Individual	Submit a draft of your REPORT RESULTS AND DISCUSSIONS section by 11:59 pm
Three weeks plus one day after your experiment	Individual	Submit INDIVIDUAL REPORT by 11:59 pm
11:59 pm on Monday, March 2	Individual	Complete 1 st half peer evaluation form
11:59 pm on Monday, April 27	Individual	Complete 2 nd half peer evaluation form

Late work policy. All laboratory work must be submitted to Canvas by the assigned due date. Late work will have 25% deducted for every 24 hours past the original due date/time. Assignments received more than 48 hours past the original due date will not be accepted.

The only exception to this policy is the case of a University-recognized excuse (where we receive a notification from Student Emergency Services). University-recognized excuses include medical issues requiring a doctor visit and/or hospitalization, religious holy days, military service, and

qualifying medical or military family leave. Please request a class absence notification from Student Emergency Services [here](#).

Regrade Requests: You have one week after a report is returned to you (i.e., posted on Canvas) to request a re-grade. Regrading requests should be submitted in writing by email to the experiment TA. The request should consist of a few sentences describing your rationale for why a re-grade is necessary for each report section requested.

Communication: The course Canvas site can be found at utexas.instructure.com. Please email the instructors through Canvas. You are responsible for ensuring that the primary email address you have recorded with the university is the one you will check for course communications because that is the email address that Canvas uses.

University Policies and Resources: For a list of important university policies and helpful resources that you may need as you engage with and navigate your courses and the university, see the [University Policies and Resources Students Canvas](#) page. The page includes the language of the University Honor Code, Title IX legal requirements for Texas employees, and information about how to receive support through the office of Disability & Access.

Academic Integrity: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the University is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community. The University of Texas has an honor code, and you are expected to abide by it. You may find a copy of it at: <http://deanofstudents.utexas.edu/conduct/>

Using someone else's work in your own writing without giving proper credit is considered plagiarism, a serious form of academic dishonesty that can result in severe penalties. Copying someone else's work, buying a paper and submitting it as your own, copying and pasting text (even with changes), or borrowing images from an online source, are some examples of plagiarism. Even if you plagiarize accidentally, you can be held responsible and penalized.

In this course, a basic expectation is that the work you turn in is your own and that your actions demonstrate respect for yourself, your classmates, teaching assistants, and the instructors. Your written work should represent *your* understanding of the experiment. This means that copying all or part of another student's (or group) laboratory report is not acceptable, nor is giving a friend your completed laboratory report so that they may copy it. Please do not hesitate to ask questions of the instructor if you have any doubts as to whether any action or behavior on your part or the part of others constitutes academic dishonest.

Learning to cite sources appropriately is an important part of becoming a professional. When you are unsure about citation, you are encouraged to **ask us** what is appropriate in the context of your assignment. Consultants at The [University Writing Center](#) can also help you determine whether you are citing sources correctly—and they have helpful guides online for using [direct quotations](#) and [paraphrasing](#). Reviewing those skills will help you feel confident that you are handling sources professionally in your writing. You can read the University's definition of plagiarism and other forms of academic dishonesty in Sec. 11–402 of the [Student Conduct Code](#). For more

information, visit the [Dean of Students' site](#).

All assignments in this course may be processed by TurnItIn, a tool that compares submitted material to an archived database of published work to check for potential plagiarism. Other methods may also be used to determine if a paper is the student's original work. Regardless of the results of any TurnItIn submission, the faculty member will make the final determination as to whether or not a paper has been plagiarized.

No use of generative AI tools permitted: This course assumes that work submitted by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as ChatGPT).

Students with Disabilities: The university is committed to creating an accessible and inclusive learning environment for students with disabilities consistent with university policy and federal and state law. Please let us know if you experience any barriers to learning so we can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A's website for contact and more information: <http://disability.utexas.edu/>. If you are already registered with D&A , please deliver your accommodation letter to us as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have our explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. We are well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.